1.3 Our Work

Manuscript-ready text, followed by three alternative flowchart layouts.

We develop a quality-aware routing and carbon-aware scheduling framework for SWITCH. Figure X summarizes four linked stages of our work.

First, we combine public benchmarks, workload traces, energy evidence and grid-carbon data to evaluate response quality, inference cost, latency and emissions. Pre-inference quality prediction and post-response assessment provide complementary evidence for comparing routing policies.

Second, we compare direct and sequential-cascade policies under quality and cost constraints. The selected Pre-LightGBM router uses task-specific finite-sample risk guards and fallback to the more capable model; post-response escalation is not activated in the selected policy.

Third, we keep model choices fixed and formulate regional allocation as a Spatial MILP, with execution evaluated through discrete-event simulation. Comparisons with Greedy and temporal scheduling alternatives examine the trade-offs among carbon emissions, tail latency and service violations.

Finally, we assess robustness under task-mix shifts, prediction noise and demand growth, and examine fairness across request categories, rebound effects and total facility emissions. Together with sensitivity and overhead analyses, these results identify the conditions under which additional routing or scheduling complexity is worthwhile and inform recommendations to SWITCH.

Suggested figure caption

Overview of our framework: performance evaluation, protected model routing, carbon-aware scheduling, and robustness and sustainability assessment.

Compact alternative

We organize our work into four linked stages (Figure X). First, we establish a common framework for quality estimation and cost, latency and carbon accounting. Second, we compare direct and cascade policies and select a Pre-LightGBM router protected by task-specific risk guards and fallback. Third, with model choices fixed, Spatial MILP and discrete-event simulation evaluate regional allocation against Greedy and temporal scheduling alternatives. Finally, stress tests, task-level fairness assessment and rebound analysis examine service risks and total environmental impact. Sensitivity and overhead analyses help identify when additional complexity is worthwhile and support recommendations to SWITCH.

A | Task dependency roadmap

A sequential task map, inspired by the dependency structure of the 2024 reference.

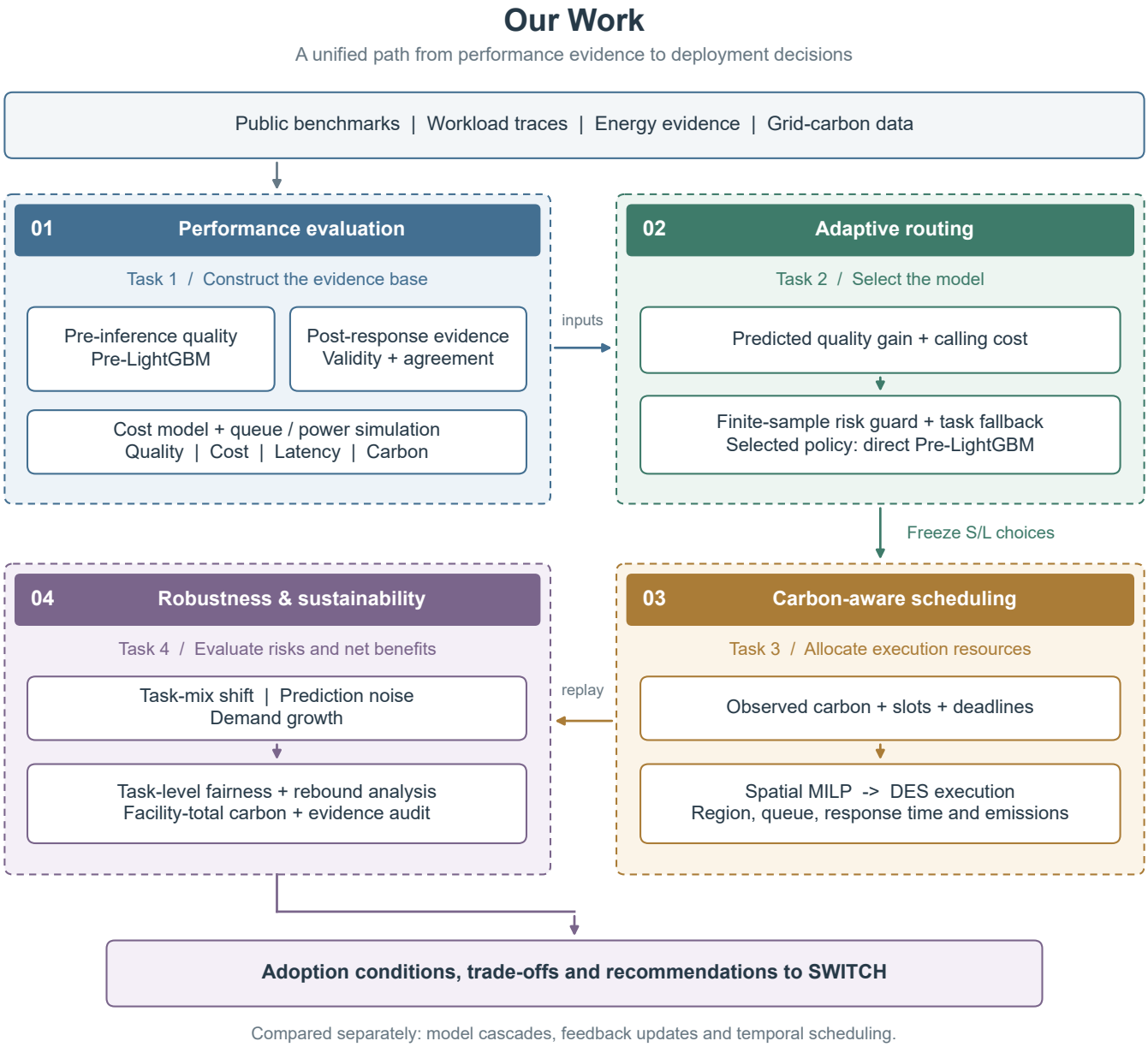


Figure X. Overview of our work.

Reading order: Task 1 to Task 2, then down to Task 3 and left to Task 4. The arrows distinguish model selection, execution scheduling and final assessment.

Use this version when the paper can allocate more vertical space to explain how the four tasks depend on one another.

All figures use the same evidence boundary: historical replay and parameterized simulation. The selected router does not activate post-response escalation; temporal policies are comparisons rather than an assumed source of savings.

B | Four-column framework

A compact overview, inspired by the parallel columns of the 2025 reference.

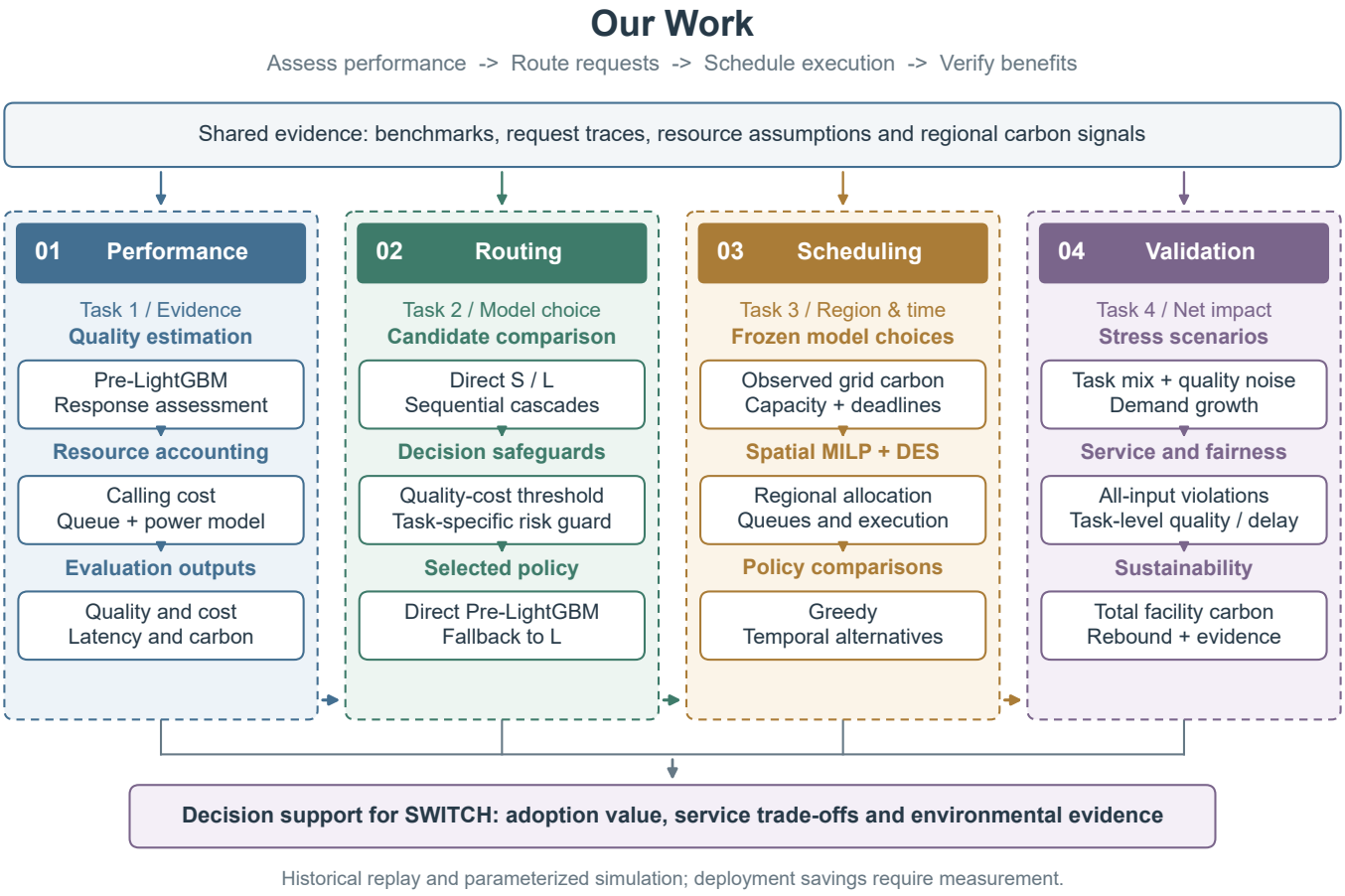


Figure X. Overview of our work.

Four colored columns separate performance evidence, routing, scheduling and validation. The bottom band collects the resulting decision support for SWITCH.

Recommended for the main manuscript: it offers the clearest balance between coverage, compactness and consistency with a conventional Our Work section.

All figures use the same evidence boundary: historical replay and parameterized simulation. The selected router does not activate post-response escalation; temporal policies are comparisons rather than an assumed source of savings.

C | Core and validation

A modular architecture, inspired by the grouped panels of the 2023 reference.

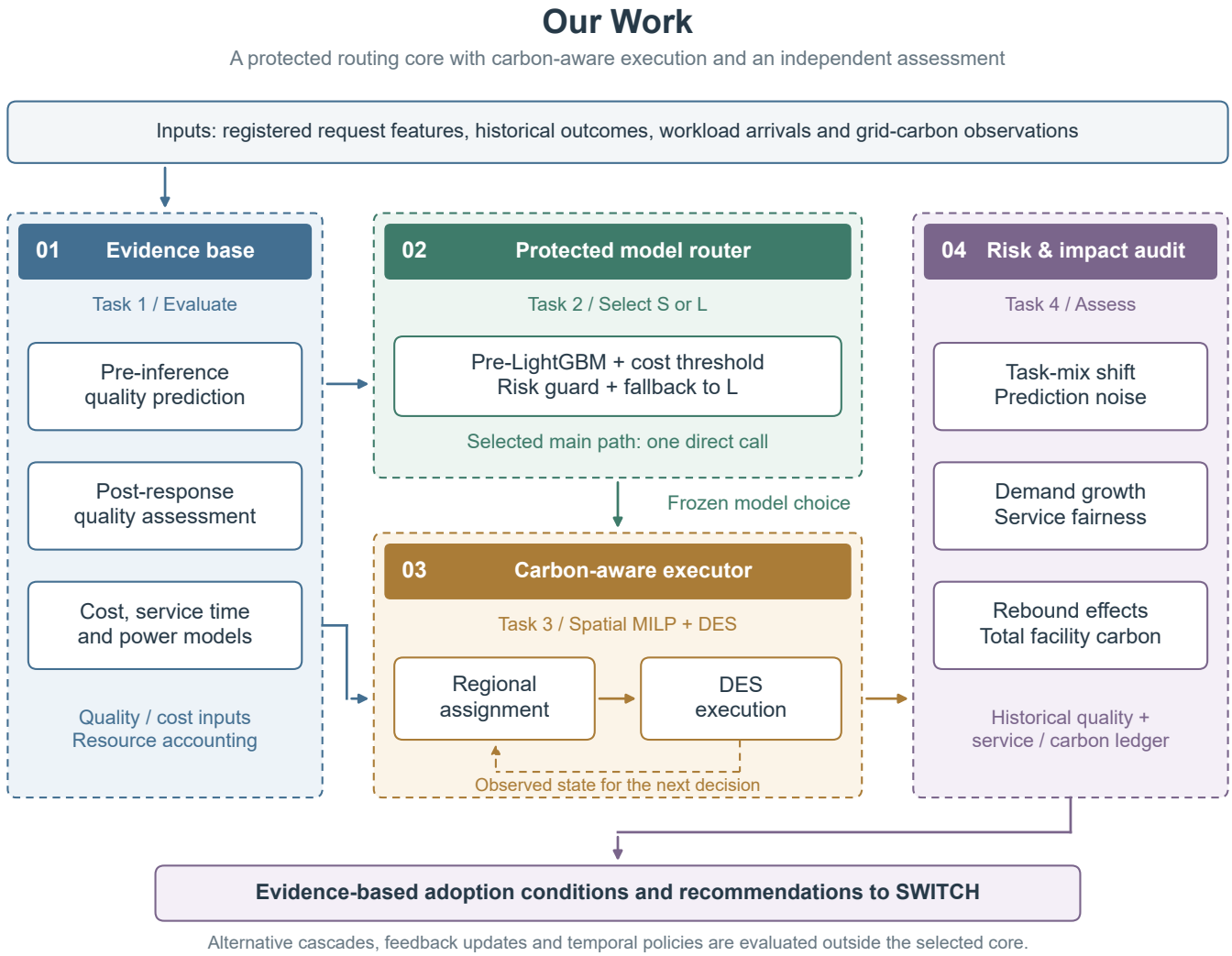


Figure X. Overview of our work.

The center highlights the selected routing and scheduling core. Evidence sits to the left, and stress testing and environmental assessment sit to the right. The dashed loop is observed scheduling-state feedback.

Use this version when the introduction should emphasize the technical contribution and distinguish the execution core from its independent assessment.

All figures use the same evidence boundary: historical replay and parameterized simulation. The selected router does not activate post-response escalation; temporal policies are comparisons rather than an assumed source of savings.